

Developer Workflow, IT Structure, Git Flow, Skills and Hosting Guide

Real-Time QR Lead Generation and Company Profile System

Document Control	Detail
Document Type	Developer Implementation Guide
System	Customizable QR-based company profile and real-time lead generation platform
Recommended MVP Stack	Next.js App Router + TypeScript + Tailwind CSS + Supabase + Vercel
Status	Developer-ready planning guide
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Senior requirement engineering position

This guide translates the SRS into a buildable developer roadmap: who builds what, how the IT structure should look, how Git should be controlled, what skills are required, and what hosting budget should be expected for pilot and production.

Core production principle: keep the first version small, stable, secure, and easy to demonstrate. The product wins when the QR scan -> landing page -> form -> saved lead -> real-time notification journey works every time.

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1. Development Objective and MVP Boundaries

The system shall be implemented as a reusable QR lead generation platform where each company has its own configurable public landing page, permanent QR link, replaceable profile content, and lead capture form. Company-level content can change, but the fixed lead receiver is controlled only by the system owner or super admin.

MVP success definition

A visitor scans a QR code, opens the correct company landing page, watches or skips the video, submits the form, the lead is saved first, then the fixed receiver is notified in real time. Admin and permitted company users see the new lead in the dashboard within the required real-time window.

The first release should not try to become a full CRM. It should be a reliable lead capture and delivery system. CRM-style pipelines, WhatsApp Business API, subscription billing, analytics, and multiple landing page templates belong to later versions.

MVP must include

- Admin login, logout, password reset, session timeout, and secure dashboard access.
- Company profile creation, edit, replacement, deactivation, and permanent QR code generation.
- Mobile-first public landing page using company logo, name, brand color, video URL, services, contact buttons, and lead form.
- Lead capture with client-side and server-side validation, duplicate protection, spam/rate-limit logic, and thank-you state.
- Lead database persistence before notification delivery.
- Real-time lead display in admin/company dashboard and email notification to the fixed receiver.
- Telegram notification as optional if configured by super admin.
- Audit log, notification log, CSV export, RBAC, and locked receiver configuration.

MVP should exclude

- Full CRM pipeline beyond basic lead status.
- Payment or subscription billing for company accounts.
- WhatsApp Business API integration, unless separately budgeted.
- Company self-registration, multi-template marketplace, bulk QR print export, and API productization.
- Direct video hosting unless Cloudinary/Supabase Storage costs and moderation process are approved. Use YouTube/Vimeo URLs first.

2. Requirements-to-Implementation Map

SRS Area	Developer Interpretation	Implementation Output
Authentication	Admin and company reps must authenticate; public landing pages remain open.	Supabase Auth, protected dashboard routes, session timeout, logout logic, password reset.
Company profile management	Company data is configurable and replaceable without changing QR identifier.	companies table, profile editor, file upload for logo, active/deactivated status.
Fixed lead receiver	Receiver config is not normal company content; it is system-level and privileged.	receiver_settings table or admin-only fields; protected UI and RLS/API enforcement.
QR management	QR points to permanent URL: /c/[company_identifier].	qrcode npm generation, PNG download, copy URL action, error correction level H.
Landing page	Dynamic, mobile-first, brandable, video-tolerant page.	Next.js dynamic route, server fetch profile, fallback for missing/broken video.
Lead form	Public submission with strong validation and anti-spam rules.	React Hook Form + Zod, server validation, rate limits, duplicate guard.
Real-time delivery	Save first, notify second; dashboard updates within 5 seconds.	Supabase Realtime subscriptions, notification queue/retry log.
Admin dashboard	Admin sees all companies/leads; company rep sees only own data.	Role-based pages, filters, search, CSV export, audit trail.

SRS Area	Developer Interpretation	Implementation Output
Security/privacy	PII access must be isolated and logged.	RLS policies, HTTPS, HttpOnly/Secure cookies, input sanitization, hashed/anonymized IP.

3. End-to-End User and System Workflow

3.1 Primary public visitor flow

Step	User/System Action	System Responsibility
1	Visitor scans QR code.	Resolve URL /c/[company_identifier] and load active company profile.
2	Landing page opens.	Display logo, company name, video area, description, services, contact buttons, and lead form.
3	Visitor submits contact form.	Validate input client-side, then submit to server endpoint for authoritative validation.
4	Server receives lead.	Apply rate limits, duplicate guard, sanitize input, hash/anonymize IP, save lead to database.
5	Lead saved.	Create notification log and trigger email/Telegram to fixed receiver. Lead saving must not depend on notification success.
6	Dashboard receives event.	Supabase Realtime pushes the new lead to admin/company rep dashboards within 5 seconds.
7	Receiver follows up.	Receiver sees lead via email/Telegram/dashboard and starts customer follow-up.

3.2 Admin operational flow

Admin login

-> create company profile

-> upload logo and configure brand color/content/video URL

-> generate permanent QR URL and PNG

-> configure fixed lead receiver from protected receiver settings

-> test QR landing page

-> monitor leads, export CSV, review audit and notification logs

3.3 Company representative flow

Company rep login

-> view only assigned company leads

-> filter/search leads

-> mark follow-up status if enabled

-> view QR code/link if permitted

-> cannot change fixed lead receiver, cannot see other companies, cannot deactivate company

4. Recommended IT Structure and Architecture

The best MVP structure is one controlled Next.js application connected to Supabase. Do not split into separate frontend and backend repositories at the beginning unless the team is already strong in distributed systems. One well-organized repo is faster, cheaper, and easier to deploy.

Layer	Recommended Technology	Responsibility
Frontend	Next.js App Router + TypeScript	Public landing pages, admin/company dashboards, routing, SEO-ready public pages.
UI	Tailwind CSS + shadcn/ui style components + Lucide icons	Mobile-first landing page and clean dashboard interface.

Layer	Recommended Technology	Responsibility
Backend/API	Next.js Route Handlers / Server Actions	Lead submission, company management, QR actions, notification trigger endpoints.
Database	Supabase PostgreSQL	Companies, leads, users, QR links, audit logs, notification logs, receiver settings.
Auth	Supabase Auth	Admin/company rep login, session management, password reset, role association.
Realtime	Supabase Realtime	Dashboard lead updates within 5 seconds.
Storage	Supabase Storage	Company logos and approved media assets. Keep video external at first.
QR	qrcode npm package	Generate PNG QR images and permanent URLs.
Notifications	Resend or SMTP for email; Telegram Bot optional	Lead notification to fixed receiver after successful DB save.
Hosting	Vercel + Supabase	Deploy app, manage env variables, SSL, preview deployments, database backend.

4.1 Environment structure

Environment	Purpose	Rules
Local	Developer machine	Uses local env file, Supabase dev project or local Supabase CLI. No production data.
Development	Shared integration testing	Connected to dev Supabase project. Used for feature demos before staging.
Staging	Client/internal acceptance testing	Production-like config, test domain, seeded sample data, no real visitor PII.
Production	Live QR platform	Protected branch deployments only from main. Real credentials and backups enabled.

4.2 Recommended environment variables

```
NEXT_PUBLIC_APP_URL=https://yourdomain.com

NEXT_PUBLIC_SUPABASE_URL=

NEXT_PUBLIC_SUPABASE_ANON_KEY=

SUPABASE_SERVICE_ROLE_KEY=

SUPABASE_JWT_SECRET=

EMAIL_PROVIDER=resend

RESEND_API_KEY=

EMAIL_FROM=Leads

TELEGRAM_BOT_TOKEN=

DEFAULT_TELEGRAM_CHAT_ID=

IP_HASH_SECRET=

RATE_LIMIT_SECRET=

CSV_EXPORT_MAX_ROWS=50000
```

5. Database, RBAC, and Supabase Structure

The data model must protect the most sensitive rule in the SRS: company content is replaceable, but the lead receiver is fixed and only super-admin controlled. Model it separately and enforce it in API and RLS policies.

Table	Important Fields	Notes
companies	id, identifier, name, logo_url, brand_color, description, services, video_url, contacts, active	identifier never changes; QR URL depends on it.

Table	Important Fields	Notes
lead_receivers	id, company_id, receiver_name, email, telegram_chat_id, active, updated_by	Super-admin only. Never editable from company profile screen.
leads	id, company_id, full_name, phone, email, organization, service, message, preferred_contact_method, status, ip_hash, created_at	Saved before notifications are attempted.
qr_codes	id, company_id, url, image_path, error_correction, created_at	Can be regenerated visually, but URL must remain permanent.
profiles/users	id, email, role, company_id, status, last_login	Role values: super_admin, admin, company_rep.
notification_logs	id, lead_id, channel, destination_hash, status, attempt_count, last_error, created_at	Tracks retries and failed notification flag.
audit_logs	id, actor_user_id, action, entity_type, entity_id, details, created_at	Required for admin actions and access/security monitoring.
landing_page_visits	id, company_id, ip_hash, user_agent_summary, created_at	Optional in v1. Useful for future QR analytics.

5.1 RBAC rules that must be enforced server-side

- Super admin: full platform access including fixed receiver settings.
- Admin: company management, leads, exports, audit logs, but fixed receiver editing only if explicitly granted.
- Company rep: own company leads only; no cross-company access; no receiver settings; export only if granted.
- Public visitor: no account; can only read active landing page content and submit lead form.
- All sensitive reads and writes must be checked in the server/API layer; hiding buttons in the UI is not security.

5.2 RLS policy mindset

Use Supabase Row Level Security on tables containing PII. For company reps, policies should filter by profile.company_id. For public lead insertion, allow insert through a controlled endpoint or RPC, not open unrestricted table inserts. Service-role operations must stay server-side only.

6. Application Route and API Structure

Route	Type	Purpose
/	Public	Marketing or redirect page for the platform owner.
/c/[identifier]	Public dynamic page	Company landing page opened by QR scan.
/c/[identifier]/not-available	Public	Friendly page for deactivated company QR scans.
/login	Public/auth	Admin and company rep login.
/admin	Protected	Admin dashboard summary.
/admin/companies	Protected	Company list, create, edit, deactivate.
/admin/companies/[id]/qr	Protected	View/copy/download QR code.
/admin/leads	Protected	Search, filter, status update, CSV export.
/admin/settings/receiver	Super-admin protected	Fixed lead receiver configuration.
/company/leads	Company protected	Company rep lead view.
/api/v1/leads	POST	Public lead submission endpoint with validation and rate limiting.
/api/v1/companies	Admin API	Company CRUD operations.
/api/v1/qr/[companyId]	Admin API	QR generation/download.
/api/v1/exports/leads	Protected API	CSV export with permission checks.
/api/v1/notifications/retry	Admin API	Retry failed notification.

6.1 Suggested repo structure

```
real-time-qr-leads/
```

```
app/

  (public)/c/[identifier]/page.tsx

  (auth)/login/page.tsx

  admin/companies/page.tsx

  admin/leads/page.tsx

  admin/settings/receiver/page.tsx

  company/leads/page.tsx

  api/v1/leads/route.ts

  api/v1/companies/route.ts

  api/v1/qr/[companyId]/route.ts

  api/v1/exports/leads/route.ts

components/

  landing/

  dashboard/

  forms/

  ui/

lib/

  supabase/client.ts

  supabase/server.ts

  auth/permissions.ts

  validation/schemas.ts

  notifications/email.ts

  notifications/telegram.ts

  qr/generate.ts

  security/rate-limit.ts

  security/ip-hash.ts

database/

  migrations/

  seed.sql

  rls-policies.sql

tests/

  unit/

  integration/

  e2e/

docs/

  api.md

  deployment.md

  data-protection.md
```

7. Developer Workflow - Step-by-Step Build Plan

The correct build order is foundation first, then public flow, then dashboards, then security hardening. Do not start with pretty dashboard charts before the lead pipeline works. That is painting the car before the engine exists.

Phase	Duration	Main Deliverables	Exit Gate
0. Setup	1-2 days	GitHub repo, Vercel project, Supabase dev project, env variables, code style, issue board.	App deploys a basic page to Vercel preview.
1. Data foundation	3-4 days	Database tables, migrations, seed data, roles, RLS draft policies.	Seed company can be loaded by identifier.
2. Auth and dashboards shell	3-5 days	Login, protected routes, layouts, role-based navigation.	Admin and company rep land on correct dashboards.
3. Company profile and QR	4-6 days	Company CRUD, logo upload, permanent identifier, QR PNG generation/download.	Updating company content does not change QR URL.
4. Public landing page	4-6 days	Dynamic landing page, video embed/fallback, contact buttons, brand color.	QR opens correct mobile-first public page.
5. Lead capture	5-7 days	Form, Zod validation, server validation, duplicate/rate-limit logic, thank-you state.	Lead is saved correctly and duplicate rules work.
6. Realtime and notifications	5-7 days	Supabase Realtime, email notification, optional Telegram, retry logic, notification logs.	Dashboard updates and fixed receiver gets notification within target window.
7. Admin lead operations	4-6 days	Search, filter, status, CSV export, audit log, retry failed notification.	Admin can manage operational leads safely.
8. Hardening and QA	5-8 days	Security checks, RLS tests, load tests, Lighthouse, broken video, deactivated QR page.	Acceptance criteria passed.
9. Launch	2-3 days	Production env, domain, SSL, privacy notice, backup check, handover docs.	Pilot company can go live.

7.1 Daily developer routine

- Pull latest develop branch before starting work.
- Create a small feature branch tied to one task or issue.
- Write or update validation schemas before building forms.
- Run lint, type check, unit tests, and relevant e2e tests locally before PR.
- Open a pull request with screenshots or screen recordings for UI changes.
- Never commit .env files, service role keys, database passwords, or API tokens.
- Update docs when API, database, environment variables, or deployment steps change.

8. Git Flow, Branch Rules, and Pull Request Discipline

Use a simple Git flow. The team should not create ceremony for the sake of ceremony. The target is clean traceability: every change has an issue, branch, PR, review, and test evidence.

Branch	Purpose	Merge Rule
main	Production-ready code only.	Only release/hotfix branches merge here after approval.
develop	Integration branch for accepted features.	Feature/fix branches merge here after PR review.
feature/-short-name	New functional work.	PR into develop.
fix/-short-name	Bug fix before release.	PR into develop.
release/v1.0.0	Final hardening for a production release.	PR into main, then back-merge/tag into develop.
hotfix/v1.0.1	Urgent production fix.	Branch from main, merge into main and develop.

8.1 Commit style

```
feat(company): add profile editor with logo upload

feat(qr): generate permanent PNG QR code
```



```
fix(leads): prevent duplicate submissions within 24 hours

security(auth): enforce API-level company access checks

chore(db): add notification log migration

docs(deploy): update Vercel environment variable guide
```

8.2 Pull request checklist

Checkpoint	Required Before Merge
Scope	PR solves one issue or one coherent feature, not five unrelated changes.
Tests	Lint, TypeScript, unit/integration/e2e tests pass as applicable.
Security	No secrets, no service-role key on client, API-level permission checks included.
Database	Migrations are reviewed; RLS impact is explained.
UI	Screenshots/screen recording provided for visible changes.
Performance	No heavy client-side queries or unindexed lead filters.
Docs	Env vars, API changes, deployment notes, and setup instructions updated.

8.3 Release versioning

Use semantic versioning: v1.0.0 for first accepted production release, v1.0.1 for hotfixes, v1.1.0 for compatible feature additions, and v2.0.0 for major behavior or data model changes.

9. Testing, QA, Security, and Definition of Done

Testing must follow the SRS acceptance criteria, especially QR permanence, fixed receiver isolation, real-time delivery, RBAC, and input security. These are not nice-to-have items. They are the spine of the product.

Test Type	Tools	Minimum Coverage
Unit tests	Vitest/Jest	Validation schemas, permission helpers, QR URL builder, IP hash helper.
Integration tests	Vitest + Supabase test project	Lead submission, notification log, duplicate guard, RLS-like server permission checks.
End-to-end tests	Playwright	QR landing page, form submit, admin login, dashboard lead display, export CSV.
Load tests	k6 or JMeter	100 concurrent visitors, form response under target, dashboard query under target.
Security tests	OWASP ZAP + manual checks	XSS/SQLi/CSRF/rate-limit/session handling.
Performance tests	Lighthouse / WebPageTest	Mobile landing page load under 3 seconds on target conditions.

9.1 Definition of Done

- Functional requirement implemented and traceable to a ticket or SRS section.
- API-level authorization verified; UI hiding alone is not accepted.
- Client-side and server-side validation both implemented where inputs exist.
- Error states are visible, specific, and friendly to the user.
- No secrets committed; environment variables documented.
- Indexes added for lead search/filter fields before production data grows.
- Feature works in staging and has passed acceptance test before production deployment.

9.2 Security must-pass list

- HTTPS enforced in production; HTTP redirects to HTTPS.
- Auth cookies/tokens are Secure and HttpOnly where applicable.
- Login and lead submission endpoints have rate limits.
- PII access is logged and isolated per company.

- Raw IP addresses are not persisted; store hashed/anonymized IP only.
- Fixed receiver change requires super-admin permission and creates audit log.
- Company rep cross-company access attempt returns 403 and is logged.

10. Skills Required to Build and Maintain the System

Skill Area	What You Must Know	Why It Matters
Requirements understanding	Read SRS, identify shall/should/may, convert requirements into tasks.	Prevents building attractive but non-compliant features.
Next.js App Router	Server components, route handlers, layouts, dynamic routes, middleware.	Core application framework.
TypeScript	Types, interfaces, API response contracts, strict mode.	Reduces runtime mistakes.
Tailwind CSS	Responsive mobile-first UI, design tokens, reusable components.	Landing page must work well on phones.
Supabase PostgreSQL	Schema design, migrations, indexes, SQL relationships.	Leads and companies require reliable structured data.
Supabase Auth/RLS	Roles, sessions, row-level policies, server-side auth checks.	Company data separation and PII protection depend on it.
Supabase Realtime	Subscriptions, channel setup, event handling, fallback behavior.	Dashboard lead update requirement.
Form validation	React Hook Form, Zod, server-side validation.	Lead form must reject invalid or unsafe data.
Security basics	OWASP Top 10, CSRF, XSS, SQLi, rate limiting, secrets management.	Public forms invite abuse; security cannot be an afterthought.
Notifications	Transactional email, retry patterns, Telegram Bot API basics.	Lead delivery is the business value.
DevOps	Vercel deploys, env variables, domains, logs, monitoring, rollback.	Keeps the product alive after coding ends.
Testing	Playwright, k6, Lighthouse, OWASP ZAP basics.	Acceptance criteria require proof, not hope.

10.1 Practical learning order

- 1 Build a static landing page in Next.js and make it perfect on mobile.
- 2 Connect Supabase and display one seeded company by identifier.
- 3 Create the lead form and save leads to the database.
- 4 Add email notification after successful save.
- 5 Add authentication and protected admin/company dashboards.
- 6 Add QR generation, CSV export, RLS, audit logs, and realtime dashboard updates.
- 7 Finally harden with rate limits, duplicate prevention, security tests, and production deployment.

11. Hosting, Environment, and Cost Guide

The SRS selected Vercel for the web app and Supabase for backend services. This is a sound MVP decision because it avoids server administration while still supporting database, authentication, real-time updates, and storage. The cost can start low, but a client-facing production service should budget for paid plans once real companies and real lead data are involved.

Planning exchange rate

ETB estimates below use a rounded planning rate of 1 USD = 161 ETB. Verify the rate before purchase or client quotation because exchange rates move.

Scenario	Recommended Setup	Estimated Monthly USD	Estimated Monthly ETB	Use When
Developer/Pilot	Vercel Hobby + Supabase Free + Resend Free + Vercel preview domain	\$0-10	ETB 0-1,610	Internal demo, learning, proof of concept, no serious client dependency.
Professional MVP	Vercel Pro (\$20) + Supabase Pro (\$25) + Resend Free/Pro as needed	\$45-65	ETB 7,245-10,465	First real client, production pilot, safer backups/support posture.
Growing Production	Vercel Pro + Supabase Pro with possible usage add-ons + Resend Pro/Scale	\$80-180	ETB 12,880-28,980	Many companies, more leads, real email volume, stronger operational needs.
Enterprise/High Traffic	Vercel Pro/Enterprise + Supabase scaled compute/storage/realtime + dedicated monitoring/support	\$200+	ETB 32,200+	Large customer base, SLA expectations, heavier realtime and data workloads.

11.1 Cost components to budget

Component	Recommendation	Cost Note
Vercel	Use Hobby for internal demo; Pro for production team/client work.	Official pricing shows Hobby free and Pro at \$20/month plus additional usage.
Supabase	Use Free for dev; Pro for production because database, backups, and support matter.	Official Pro plan is \$25/month with higher limits than free plan.
Email	Start with Resend Free for low email volume; upgrade when lead notifications grow.	Resend Free includes 3,000 emails/month; Pro is \$20/month for 50,000 emails/month.
Domain/DNS/SSL	Use Cloudflare DNS/Registrar where possible; Vercel also supports custom domains.	Cloudflare Registrar uses at-cost registration/renewal and includes DNS/CDN/SSL features.
Telegram	Use Bot API as optional secondary notification channel.	Main cost is developer time and operational setup, not monthly platform fee.
Video	Use YouTube/Vimeo URLs in v1.	Avoid paying to store/stream company videos until real demand exists.
Monitoring	Start with Vercel/Supabase logs; add Sentry or similar later.	Budget later if production incidents become difficult to trace.

11.2 Hosting decision by stage

- Learning/demo stage: do not spend money unless a custom domain is needed. Use free tiers responsibly.
- First production client: pay for Vercel Pro and Supabase Pro. Cheap hosting is expensive when a real lead disappears.
- Email-heavy production: upgrade transactional email plan and monitor failed notification logs.
- Data growth stage: add indexes, archive old leads, monitor Supabase database size and realtime usage.
- High-value clients: prepare SLA, backup policy, incident response process, and formal support terms.

11.3 Production environment checklist

- Production Vercel project connected to main branch only.
- Production Supabase project separated from development and staging.
- All production secrets stored in Vercel environment variables, never in Git.
- Custom domain connected, HTTPS active, and HTTP redirect tested.
- Supabase backups checked and restore procedure documented.
- Email sending domain verified with SPF, DKIM, and DMARC where supported.
- Privacy notice available on landing pages before going live.
- Super-admin account secured; 2FA should be enabled where platform supports it.

12. Launch Checklist and Handover Package

Area	Checklist
Functional	QR opens correct page; company update does not change QR; lead saves; dashboard updates; fixed receiver notified; export works.
Security	RBAC tested; company rep cannot cross-read leads; rate limits work; HTTPS enforced; secrets protected.
Content	Company name/logo/video/contact/buttons reviewed; privacy notice visible; deactivated QR page friendly.
Operations	Admin knows how to create company, generate QR, view/export leads, retry notification, change receiver safely.
Backup	Supabase backup policy known; data export procedure documented.
Support	Known support email/Telegram/channel defined; issue severity levels agreed.
Documentation	README, deployment guide, env var list, database schema, test plan, and user admin guide completed.

12.1 Handover deliverables

- GitHub repository with clean commit history and protected main branch.
- README with local setup commands and environment variable guide.
- Database migration files and RLS policy scripts.
- Deployment guide for Vercel and Supabase.
- Admin user manual with screenshots for company setup, QR download, leads, exports, receiver settings, and notification retry.
- Test evidence showing the acceptance criteria were checked.
- Support and maintenance notes: backup, logs, incident handling, and upgrade procedure.

13. Source Notes

This guide is based on the uploaded SRS v2.1 for the Real-Time QR Lead Generation and Company Profile System. The SRS defines the core flow, user roles, functional and non-functional requirements, selected technology stack, first version scope, and acceptance criteria.

Source	Used For	Reference
Uploaded SRS v2.1	System purpose, user flow, roles, functional requirements, NFRs, selected stack, acceptance criteria.	Real_Time_qr_generation_SRS(1).pdf
Vercel Pricing	Hosting plan assumptions for web app and deployment.	https://vercel.com/pricing
Supabase Pricing	Backend/database/auth/realtime/storage plan assumptions.	https://supabase.com/pricing
Resend Pricing	Email notification plan assumptions.	https://resend.com/pricing
Cloudflare Registrar	Domain/DNS/SSL registration assumptions.	https://www.cloudflare.com/products/registrar/
Trading Economics USD/ETB	Planning exchange rate cross-check.	https://tradingeconomics.com/ethiopia/currency

Final senior engineering advice

Build the lead path first. QR, landing page, form, database save, receiver notification, dashboard update. Once that backbone is strong, every extra feature becomes an upgrade instead of a rescue mission.